

Voltage sag source location based on instantaneous energy detection

Wei Kong^{a,b,*}, Xinzhou Dong^a, Zhe Chen^b

^a Department of Electric Engineering, Tsinghua University, China

^b Institute of Energy Technology, Aalborg University, Denmark

ARTICLE INFO

Article history:

Received 30 April 2007

Received in revised form 8 January 2008

Accepted 16 March 2008

Available online 28 April 2008

Keywords:

Voltage sag

Instantaneous energy based on variation components

Hilbert transform

Voltage sag source detection

Distributed generation units

ABSTRACT

Voltage sag is a major power quality problem, which could disrupt the operation of voltage-sensitive equipment. This paper presents the method based on variation components-based instantaneous energy for voltage sag source detection. Simulations have been performed to provide the thorough analysis for system with distributed generation units. The studies show that the presented method can effectively detect the location of voltage sag source.

© 2008 Elsevier B.V. All rights reserved.

1. Introduction

Power quality has caught more attention than before. Voltage sag is a major power quality problem which may be detrimental to voltage-sensitive loads. Monitoring and detection of voltage sag are necessary to improve the power quality. Much work has been done to develop the systems that automatically identify the major types of power quality disturbances in system grid [1–3].

In order to deal with voltage sag, the information of voltage sag source location is also necessary. Some researches have been carried out. In [4], a direction finder for voltage sag is proposed; the proposed method is based on the variation of active power before and after the sag and the integral of the variation of the active power over the duration of the sag to judge the direction of the voltage sag source. In [5], the method using the slope of a line, obtained by least square fitting approach, is proposed for voltage sag direction detection where voltage and current phasors and power factor are utilized. In [6], the polarity of the real current component is proposed to determine the voltage sag location, with the presented method, voltage sag can be detected by examining the polarity of the RMS current and the power factor angle at the inception of the fault. In [7], the principle based on impedance and its angle detection before and after the sag is presented, with

the proposed approach, position of the sag source in an integrated network can be obtained from simultaneous readings of different power quality monitoring units' measured impedance. In [8], the method based on the resistance of the measured impedance is presented. The impedance estimation method is adopted to obtain the power–frequency positive-sequence resistance component at the metering point, so that the voltage sag source location can be determined by the variation of resistance.

Owing to the voltage sag is mainly resulted from the system faults; it could be benefit from the fault variation components of the system to detect the voltage sag source. This paper presents the applications of instantaneous active- and reactive-energy based on the variation components for voltage sag source detection. The section of line of the voltage sag location can be determined from the direction information of instantaneous energy of different power quality monitoring units.

The paper is structured as follows: the basic principle of the method is presented in Section 2; instantaneous active-power and energy, and reactive-power and energy based on Hilbert transform are analyzed in Section 3; Simulation studies are used to provide the thorough analysis for system with distributed generation units in Section 4; then the conclusions are drawn in Section 5.

2. Basic principles

Because system faults are the main reason resulting in the voltage sags, the voltage sag location method can be studied from the point of view of the fault analysis theory.

* Corresponding author at: Pon. 101, Aalborg University, Aalborg 9220, Denmark. Tel.: +45 96 35 92 80.

E-mail address: jxkongwei@hotmail.com (W. Kong).

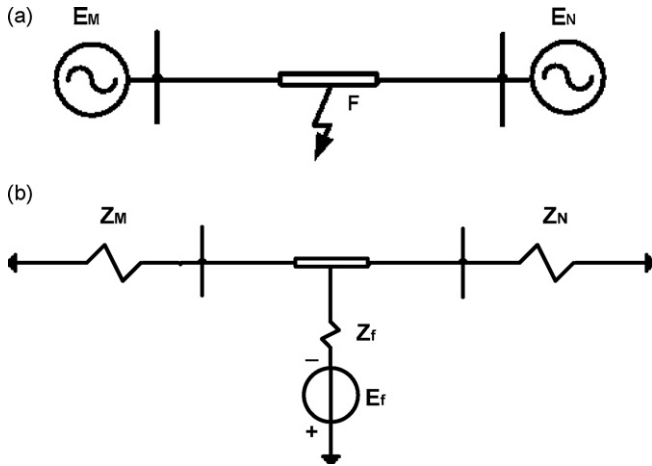


Fig. 1. The typical system and fault-component superposition circuit.

When the faults happen, the system can be divided into pre-fault circuit and the fault-component circuit according to superposition theory. Considering a typical system shown in Fig. 1(a), when the fault happens at F , the fault-component superposition circuit can be shown as Fig. 1(b), where E_f is the fault-component superposition source, and Z_f is the fault impedance, Z_M and Z_N are the equivalent impedance of the system E_M and system E_N , respectively.

The fault results in the fault-component superposition source. The fault-component superposition energy should possess the directional characteristic, which can be used to locate the fault source, so that voltage sag resulting from the system faults can be located by the fault-component energy direction.

Instantaneous energy based on fault component is composed of instantaneous active- and reactive-energy. Both of them possess the directional characteristic. The location of voltage sag can be determined according to the direction of instantaneous active- and reactive-energy. Theoretically as long as one of them is significant, the direction of the voltage sag should be distinct.

However, the distribution of instantaneous energy based on fault component is mainly affected by the equivalent impedance of the system. If the resistance of the equivalent impedance is close to zero, the value of instantaneous active-energy will be periodically close to zero. The reliability and sensitivity of the detection criterion based on instantaneous active-energy would be weakened. In this situation the value of instantaneous reactive-energy is significant, so the detection criterion based on instantaneous reactive-energy should be more reliable and higher sensitive. In this paper, both instantaneous active- and reactive-power and energy are used for voltage sag source location.

The main factors which can affect the strength of the fault-component based energy are listed as

- (1) the voltage inception angle of the time of the fault;
- (2) the fault impedance.

3. Instantaneous active- and reactive-power and energy

3.1. Instantaneous active-power

Instantaneous active-power can be obtained as follow:

$$p_i(t) = u_a(t)i_a(t) + u_b(t)i_b(t) + u_c(t)i_c(t) \quad (1)$$

where $p_i(t)$ represents instantaneous active-power. $u_a(t)$, $u_b(t)$, $u_c(t)$ and $i_a(t)$, $i_b(t)$, $i_c(t)$ represent the instantaneous values of voltage and current of three phases, respectively.

3.2. Instantaneous reactive-power based on Hilbert transform

Many researches on instantaneous reactive-power have been carried out. In [9], the vectors of instantaneous voltage and current are defined, and then instantaneous reactive power can be considered as the second asymmetrical tensor resulting from the out-product operation of voltage and current vectors. In [10], the ps - qs theory of instantaneous power based on Hilbert transform is presented, which is used for reactive power control in single-phase circuits. And in [11], the definitions of average and instantaneous

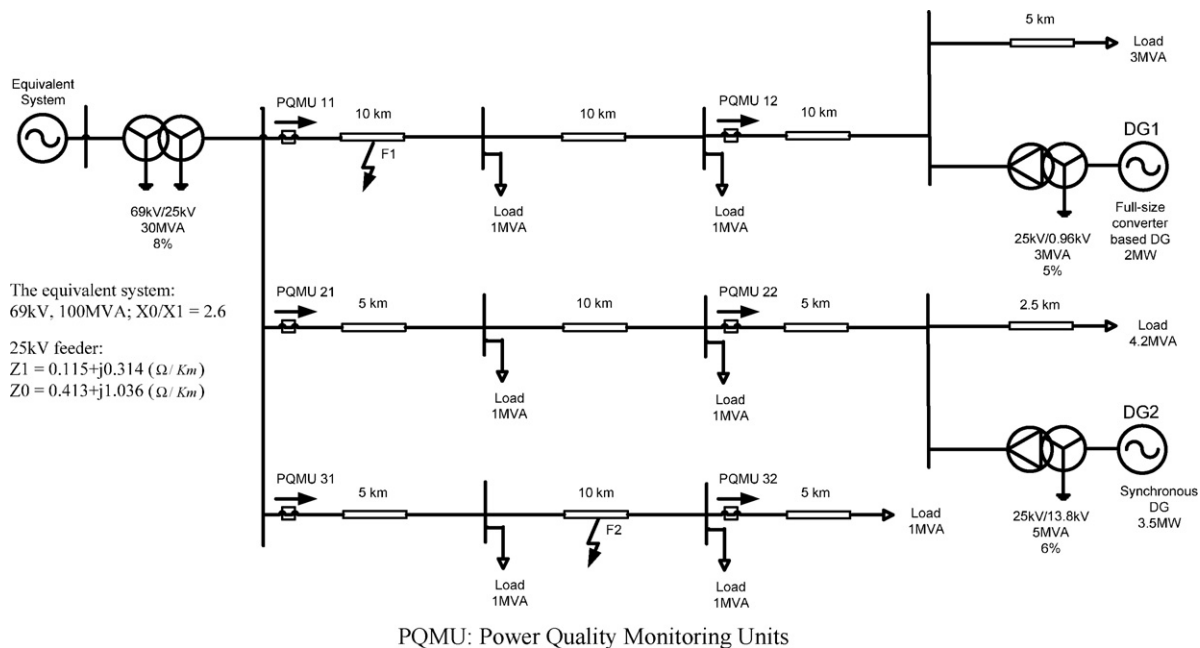


Fig. 2. Single-line diagram of a 25 kV distribution system with DGs.

reactive-power based on Hilbert transform for non-sinusoidal circuit are presented.

Most of instantaneous reactive-power theories are developed for reactive-power compensation and controller strategy development. In this paper, the directional characteristic of

the instantaneous energy is the sticking point, so the integral of instantaneous reactive-power should possess the directional characteristic. The instantaneous reactive-power based on Hilbert transform can provide with directional information [11].

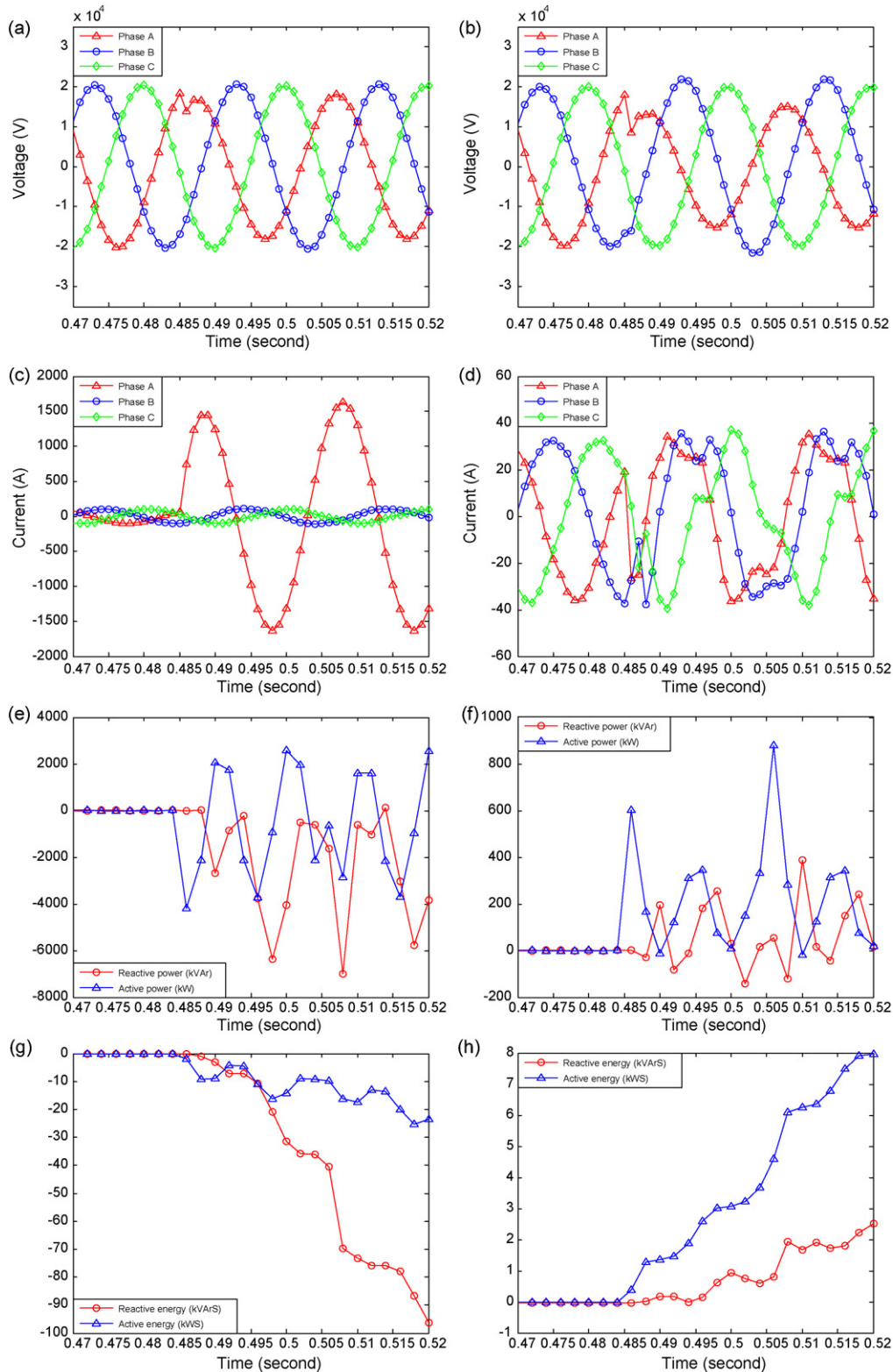


Fig. 3. Voltage and current waveforms and variations of instantaneous active- and reactive-power and energy at PQMU11 and PQMU12 (single-phase-grounding fault at F1). Voltage waveforms at (a) PQMU11 and (b) PQMU12. Current waveforms at (c) PQMU11 and (d) PQMU12. Variations of instantaneous active- and reactive-power at (e) PQMU11 and (f) PQMU12. Instantaneous active- and reactive-energy at (g) PQMU11 and (h) PQMU12.

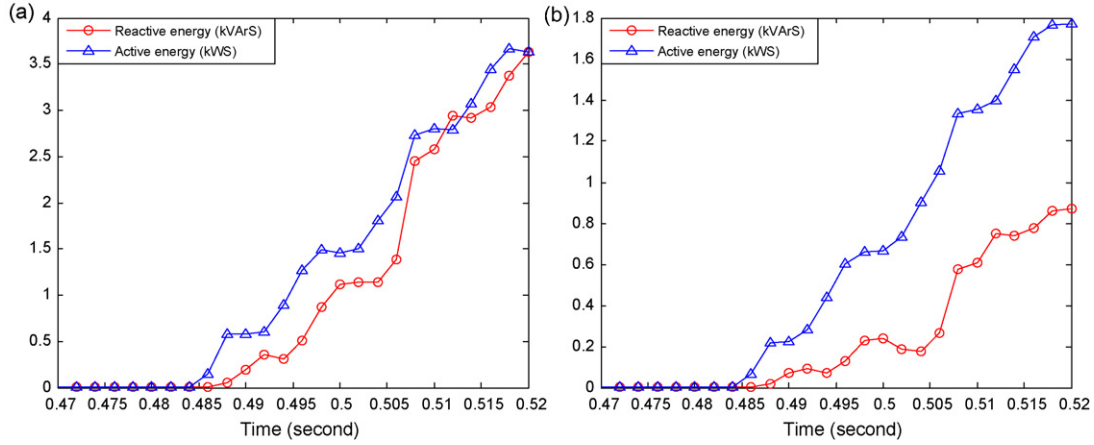


Fig. 4. Variations of instantaneous active- and reactive-power and energy at PQMU21 and PQMU31 (single-phase-grounding fault at F1). Instantaneous active- and reactive-energy at (a) PQMU21 and (b) PQMU31.

The impulse response of Hilbert transform can be described in Eq. (2):

$$h(t) = \frac{1}{\pi t}, \quad -\infty < t < \infty \quad (2)$$

Then the response of the frequency domain of Hilbert transform can be deduced:

$$H(\omega) = \begin{cases} -j, & \omega > 0 \\ j, & \omega < 0 \end{cases} \quad (3)$$

It is shown that Hilbert transform is a kind of allpass filter. The phase angle is transformed $\pm 90^\circ$ according to the value of ω and the amplitude is kept to 1.

If the instantaneous voltage is $u(t)$, the equation of Hilbert transform can be expressed as:

$$\hat{u}(t) = u(t) * h(t) = \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{u(\tau)}{t - \tau} d\tau \quad (4)$$

The current can be mapped to the $u(t) - \hat{u}(t)$ coordinate, and decomposed into two components, i_p and i_q . Obviously, two components are orthogonal. Instantaneous reactive power can be defined as:

$$q_i(t) = \hat{u}(t)i_q(t) \quad (5)$$

where $u(t)$, $\hat{u}(t)$, i_p and i_q represent instantaneous voltage, the voltage resulting from Hilbert transform, and the active- and reactive-components of the current, respectively; $q_i(t)$ represents instantaneous reactive power.

In order to realize Hilbert transform, Hilbert digital filters [12,13] should be utilized. One is designed for voltage signal, and the other one is designed for current signal. The details of the design of Hilbert transform are not discussed in this paper, which are outside the scope of this paper.

3.3. Instantaneous active- and reactive-power and energy based on variation components

The extraction of variation components can be obtained as follows:

$$\begin{cases} \Delta u_k(t) = u_k(t) - 2u_k(t - T) + u_k(t - 2T) \\ \Delta i_k(t) = i_k(t) - 2i_k(t - T) + i_k(t - 2T) \end{cases} \quad (6)$$

where $u_k(t)$ and $i_k(t)$ represent the sample values of voltage and current, respectively; $\Delta u_k(t)$ and $\Delta i_k(t)$ represent the variations

of voltage and current respectively; T represents the period of the signal.

Instantaneous active- and reactive-power and energy based on variation components can be defined as follows:

$$\Delta p_i(t) = \sum_k \Delta u_k(t) \Delta i_k(t) \quad (7)$$

$$\Delta q_i(t) = \sum_k \Delta \hat{u}_k(t) \Delta i_{qk}(t) \quad (8)$$

where $k = a, b, c$, which represent phase A, phase B and phase C, respectively; $\Delta \hat{u}_k(t)$ and $\Delta i_{qk}(t)$ represent the values of variations of voltage and current resulting from Hilbert transform, respectively. $\Delta p_i(t)$ and $\Delta q_i(t)$ represent the values of instantaneous active- and reactive-power based on variation components.

The integral of variations of instantaneous active- and reactive-power can be deduced as follows:

$$\Delta E_p(t) = \int_0^t \Delta p_i(t) dt \quad (9)$$

$$\Delta E_q(t) = \int_0^t \Delta q_i(t) dt \quad (10)$$

where $\Delta E_p(t)$ and $\Delta E_q(t)$ represent the values of instantaneous active- and reactive-energy based on variation components.

4. Simulation studies

4.1. Simulation model description

The simulations are utilized to investigate into the characteristics of the proposed method.

The single-line diagram of the simulation system is shown in Fig. 2, including two distributed generation units (DGs). DG1 is connected to the grid via full-size converter based on IGBT PWM (pulse width modulation) voltage source converter. And DG2 is a traditional synchronous generator with the excitation and governor modules. The studies are performed based on the MATLABTM software package.

Power quality monitoring units (PQMUs) are installed as indicated in Fig. 2. The faults are simulated at different situations and the results are provided with corresponding instantaneous energy direction.

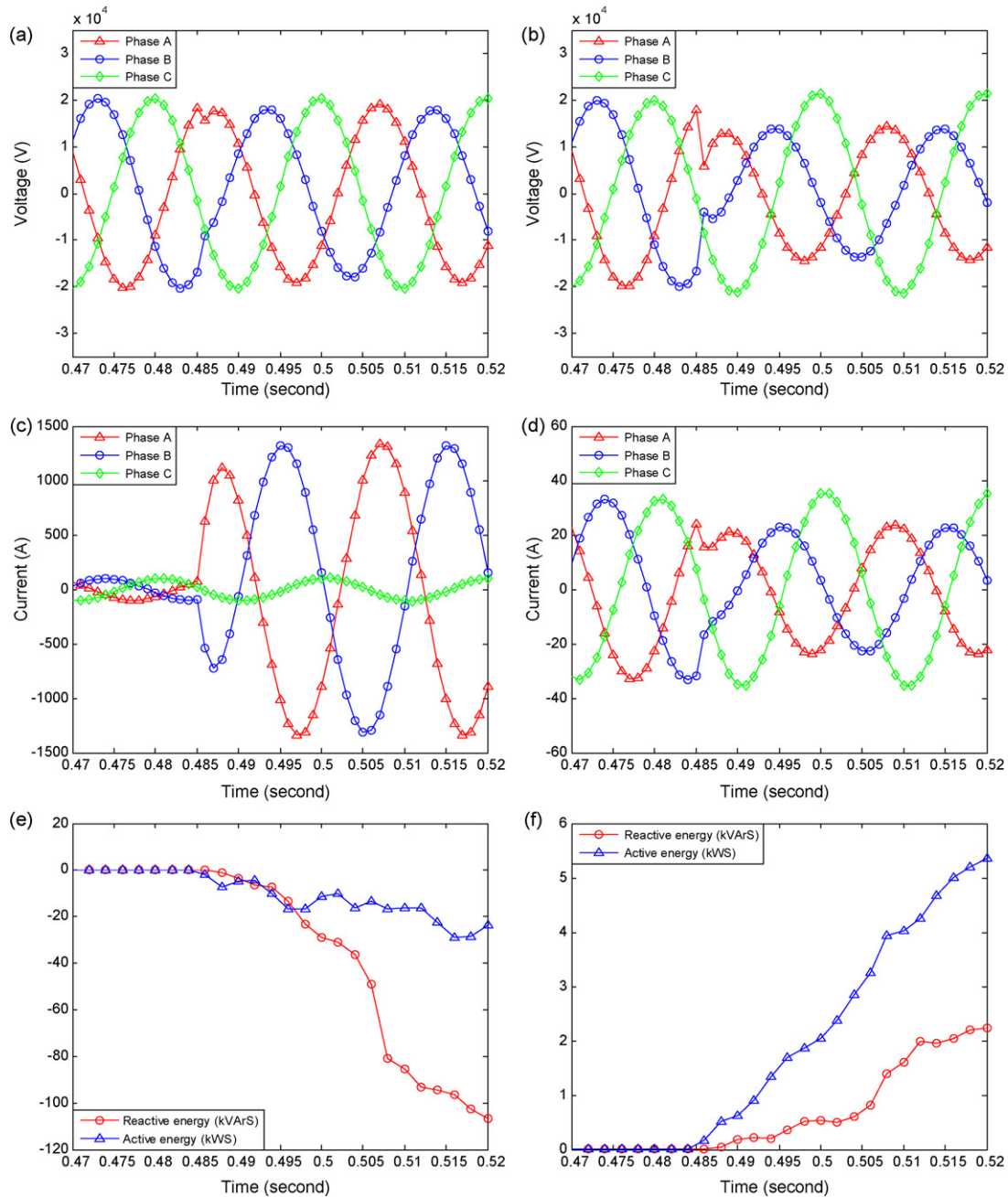


Fig. 5. Voltage and current waveforms and variations of instantaneous active- and reactive-energy at PQMU31 and PQMU32 (phase-to-phase-grounding fault at F2). Voltage waveforms at (a) PQMU31 and (b) PQMU32. Current waveforms at (c) PQMU31 and (d) PQMU32. Instantaneous active- and reactive-energy at (e) PQMU31 and (f) PQMU32.

4.2. Cases studies

The different fault types can result in different types of voltage sag. The single-phase grounding fault, phase-to-phase fault and phase-to-phase grounding fault (namely unbalanced faults) result in unbalanced voltage sag. The three-phase fault and three-phase-grounding fault (namely balanced faults) result in balanced voltage sag. In the simulation cases, the responses of the proposed method to the different types of the voltage sag are studied.

4.2.1. Unbalanced voltage sag

In this paragraph, the simulation results of the proposed instantaneous energy detection method for voltage sag source location are presented for unbalanced voltage sag situations.

There are six nodes on the main feeders where PQMUs are installed. The downstream direction of PQMU is defined from the Bus to Line as indicated in Fig. 2.

The faults located at F1 and F2 are simulated that produce voltage sag throughout the network. The faults happen in the middle of the line, and the inception of the faults is at 0.485 s. The fault impedance is 10 Ω .

(a) Single-phase-grounding fault at F1.

The voltage and current waveforms at PQMU11 and PQMU12, and the variations of instantaneous active- and reactive-power and instantaneous active- and reactive-energy are shown in Fig. 3. Instantaneous active- and reactive-energy at PQMU21 and PQMU31 are shown in Fig. 4.

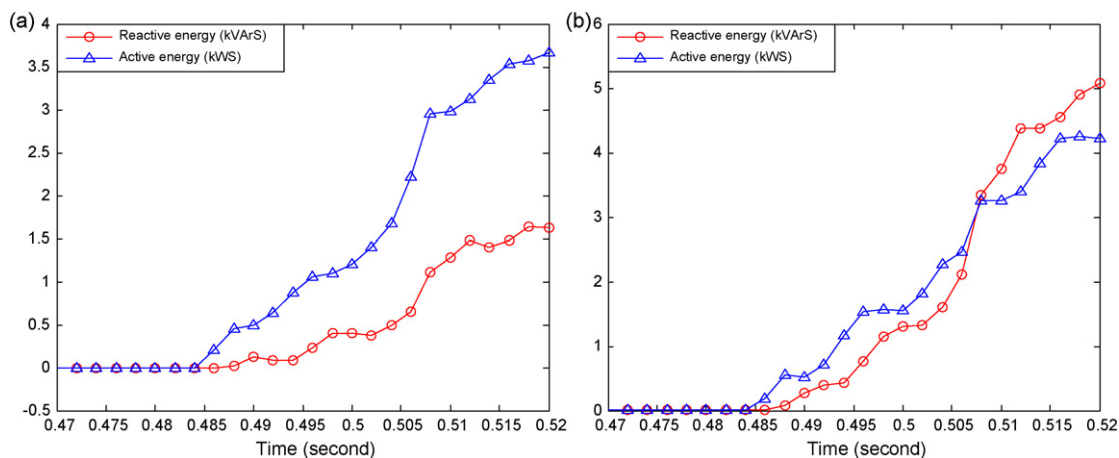
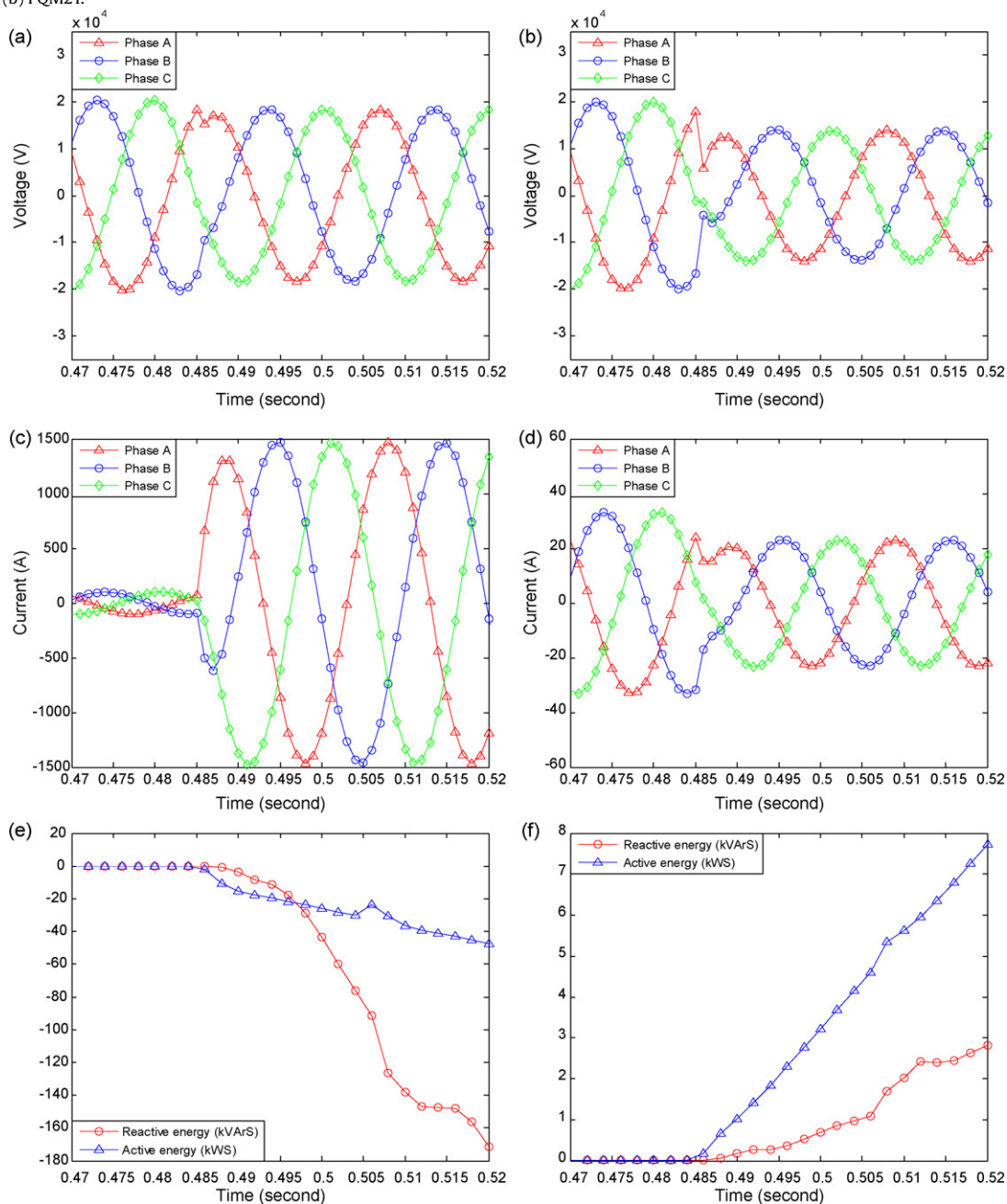


Fig. 6. Variations of instantaneous active- and reactive-energy at PQMU11 and PQMU21 (phase-to-phase-grounding fault at F2). Instantaneous active- and reactive-energy at (a) PQMU11 and (b) PQMU21.



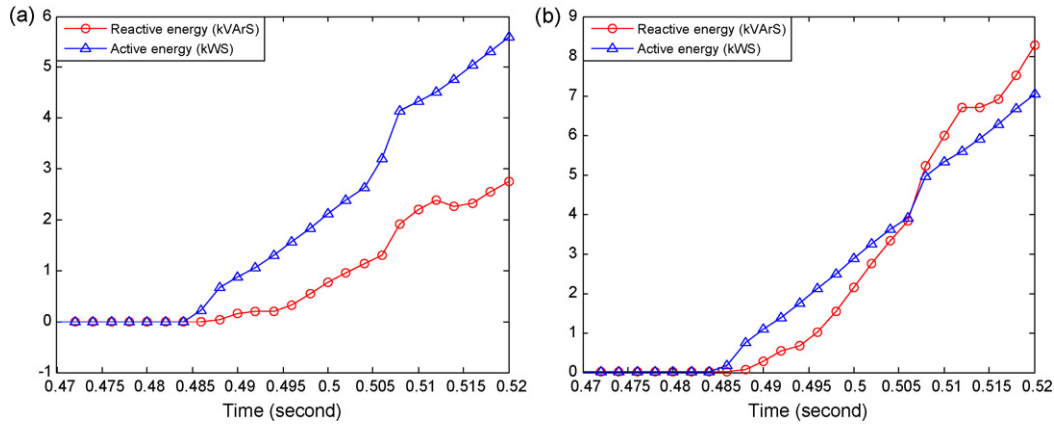


Fig. 8. Variations of instantaneous active- and reactive-power and energy at PQMU11 and PQMU21 (three-phase fault at F2). Instantaneous active- and reactive-energy at (a) PQMU11 and (b) PQMU21.

Instantaneous active- and reactive-energy of PQMU11 is negative, which indicates that the voltage sag source is downstream; and instantaneous active- and reactive-energy of PQMU12 is positive, which indicates that the voltage sag source is upstream. So the voltage sag source should locate between PQMU11 and PQMU12. And instantaneous active- and reactive-energy of PQMU21 and PQMU31 shown in Fig. 4 are both positive, which mean that the voltage sag location is behind PQMU21 and PQMU31.

In Fig. 3(e) and (f), the polarity of the first peak of instantaneous power possesses the same directional characteristic as instantaneous energy, which is coincident with the result observed by the authors in [4]. The reason of this phenomenon is that in the proposed method instantaneous active- and reactive-power are based on variation components which are resulted from the fault-component superposition source, so that the polarity of the first peak of instantaneous power should possess the definitely directional characteristic.

In Fig. 3(g) and (h), it is shown that the variations of instantaneous energy on the system side (PQMU11) is more significant than that on the DG side (PQMU12), the main reasons are that the system possesses the larger short-circuit capacity than DG system, and in this situation the full-size inverter-based DG is adopted, the response of the DG is restricted by the strategy of the controller. The inverter interface has a nonlinear impedance characteristic, which also affects the distribution of instantaneous energy.

(b) Phase-to-phase-grounding fault at F2 on single-source line.

The voltage and current waveforms at PQMU31 and PQMU32 and the variations of instantaneous active- and reactive-energy are shown in Fig. 5. Instantaneous active- and reactive-energy at PQMU11 and PQMU21 are shown in Fig. 6.

Instantaneous active- and reactive-energy of PQMU31 is negative, and instantaneous active- and reactive-energy of PQMU32 is positive, which indicate that the voltage sag source locates between PQMU31 and PQMU32. And instantaneous active- and reactive-energy of PQMU11 and PQMU21 shown in Fig. 6 are both positive, which mean that the voltage sag location is behind PQMU11 and PQMU21.

In this situation, the fault is simulated at a single-source feeder. Instantaneous energy direction can correctly indicate the voltage

sag source. It is shown that the algorithm based on instantaneous energy direction can be effective in the single-source radial topology grid as well.

The above results are for some typically unbalanced fault situations. Many cases have been studied, including different power flow condition, different monitoring zones, and fault durations. Simulation results show that the presented method can properly indicate the location of the voltage sag sources.

4.2.2. Balanced voltage sag

Three-phase faults result in balanced voltage sag. Normally three-phase faults could lead to the largest short-circuit current. The presented method is based on the variations of instantaneous active- and reactive-energy, which should possess the high sensitivity to three-phase faults. The simulation results also verify this conclusion. A three-phase fault located at F2 is simulated.

The voltage and current waveforms at PQMU31 and PQMU32 and the variations of instantaneous active- and reactive-energy are shown in Fig. 7. Instantaneous active- and reactive-energy at PQMU11 and PQMU21 are shown in Fig. 8.

Instantaneous active- and reactive-energy of PQMU31 is negative, and instantaneous active- and reactive-energy of PQMU32 is positive, which indicate that the voltage sag source locates between PQMU31 and PQMU32. And instantaneous active- and reactive-energy of PQMU11 and PQMU21 shown in Fig. 8 are both positive, which mean that the voltage sag location is behind PQMU11 and PQMU21.

4.2.3. The effects of inception angle of the faults on instantaneous energy

The most rigorous situation is that the fault happens when the voltage passes the zero. The smallest transient energy would be transmitted in this situation. The single-phase-grounding (phase A) fault located at F1 is simulated at 0.491 s.

The voltage and current waveforms at PQMU11 and PQMU12 and the variations of instantaneous active- and reactive-power and instantaneous active- and reactive-energy are shown in Fig. 9.

Instantaneous active- and reactive-energy of PQMU11 and PQMU12 indicate that the voltage sag source locates between PQMU11 and PQMU12.

Compared Fig. 9(e) and (f) with Fig. 3(e) and (f), it is shown that the inception angle of the fault will affect instantaneous power.

Fig. 7. Voltage and current waveforms and variations of instantaneous active- and reactive-energy at PQMU31 and PQMU32 (three-phase fault at F2). Voltage waveforms at (a) PQMU31 and (b) PQMU32. Current waveforms at (c) PQMU31 and (d) PQMU32. Instantaneous active- and reactive-energy at (e) PQMU31 and (f) PQMU32.

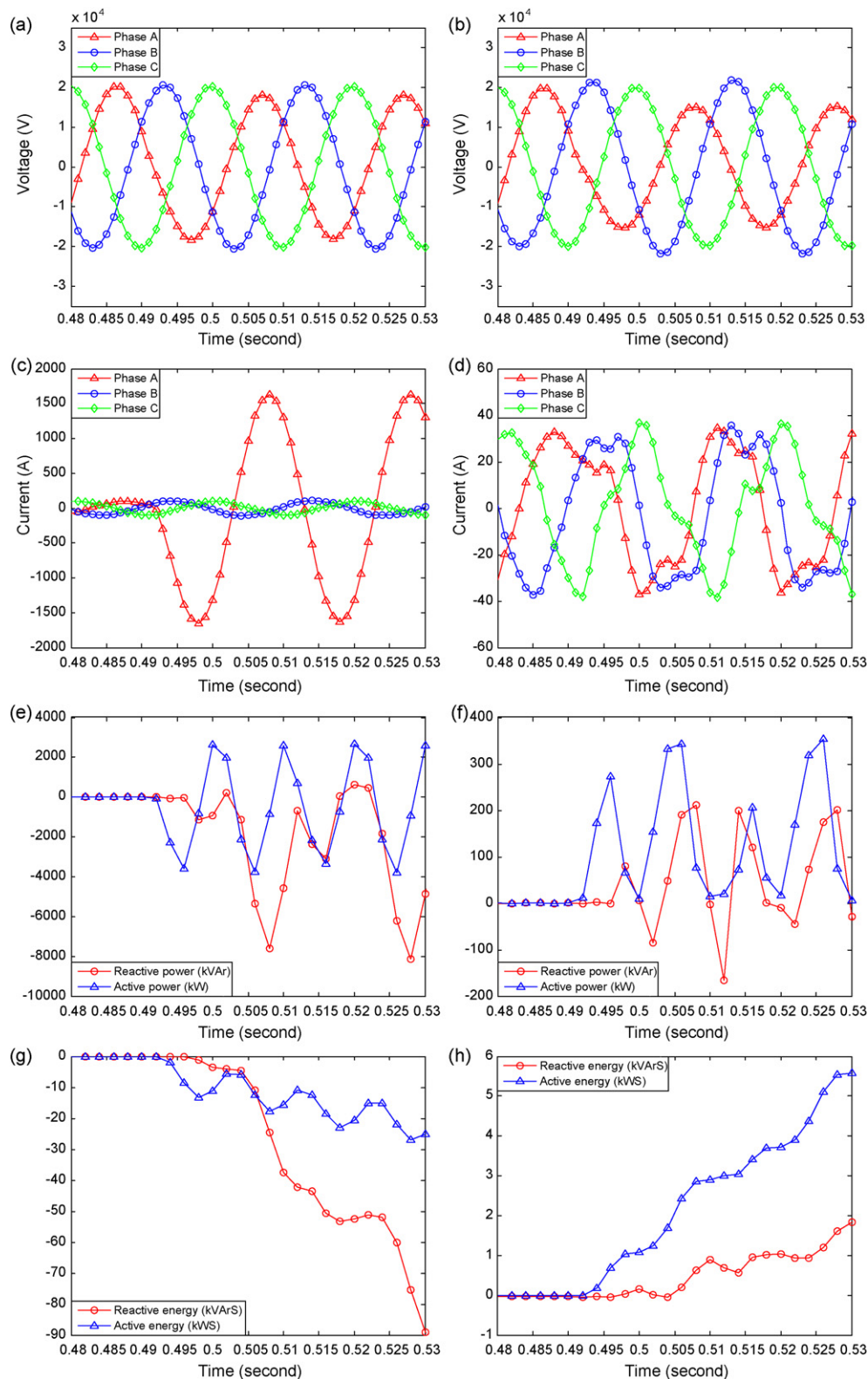


Fig. 9. Voltage and current waveforms and variations of instantaneous active- and reactive-power and energy at PQMU11 and PQMU12 (single-phase-grounding fault at F1). Voltage waveforms at (a) PQMU11 and (b) PQMU12. Current waveforms at (c) PQMU11 and (d) PQMU12. Variations of instantaneous active- and reactive-power at (e) PQMU11 and (f) PQMU12. Instantaneous active- and reactive-energy at (g) PQMU11 and (h) PQMU12.

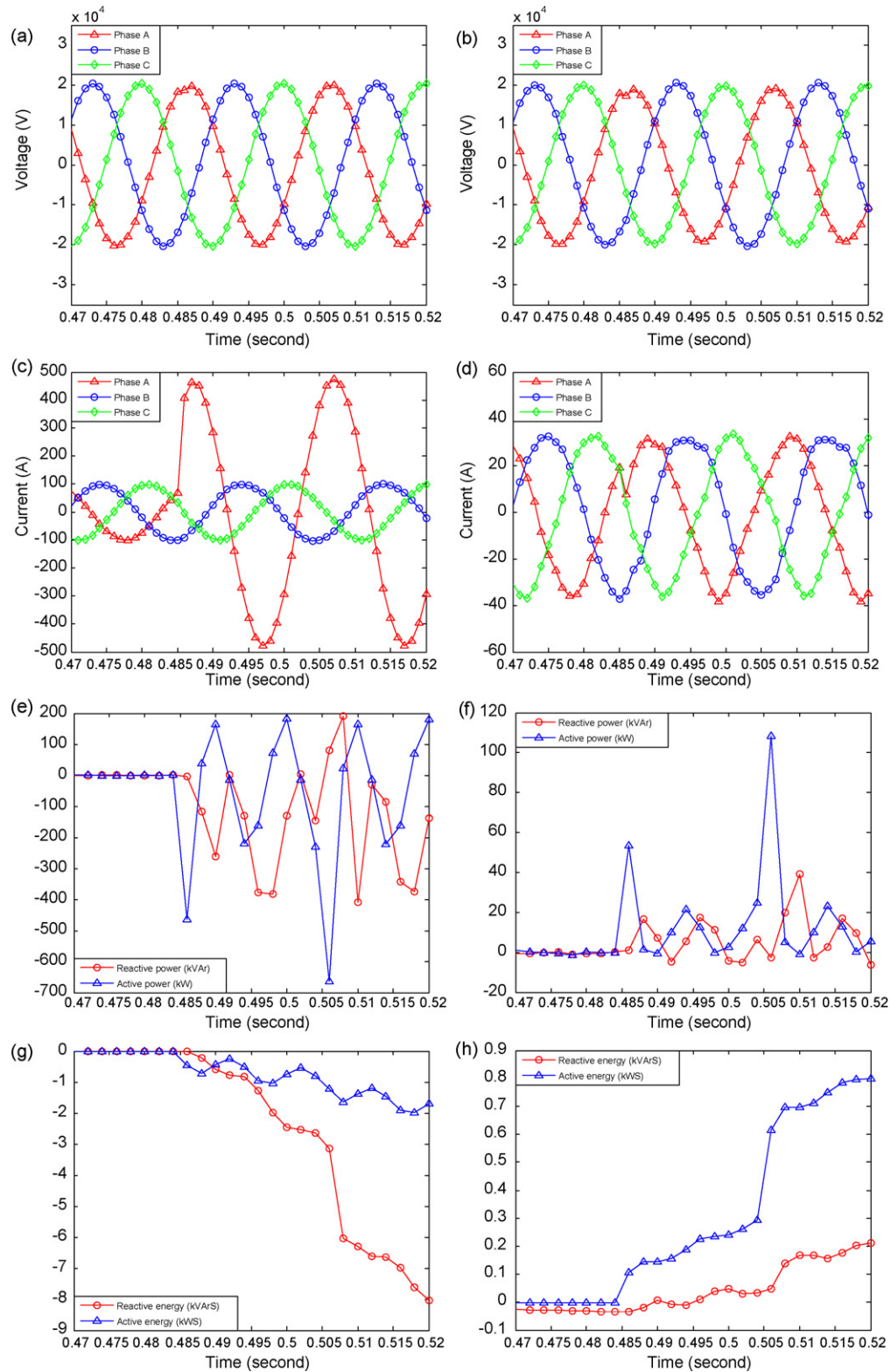


Fig. 10. Voltage and current waveforms and variations of instantaneous active- and reactive-power and energy at PQMU11 and PQMU12 (single-phase-grounding fault at F1). Voltage waveforms at (a) PQMU11 and (b) PQMU12. Current waveforms at (c) PQMU11 and (d) PQMU12. Variations of instantaneous active- and reactive-power at (e) PQMU11 and (f) PQMU12. Instantaneous active- and reactive-energy at (g) PQMU11 and (h) PQMU12.

If the faults happen when the voltage passes the zero, instantaneous energy would be smallest. In this situation the duration of the integral should be prolonged. The simulation results show that the presented method can properly indicate the location of the voltage sag sources as well.

4.2.4. The effects of fault impedance on instantaneous energy

The fault impedance could reduce instantaneous energy. The single-phase-grounding fault located at F1 is simulated. The inception of fault is at 0.485 s. The value of fault impedance is changed from 10 to 50 Ω .

The voltage and current waveforms at PQMU11 and PQMU12, and the variations of instantaneous active- and reactive-power and instantaneous active- and reactive-energy are shown in Fig. 10.

Instantaneous active- and reactive-energy of PQMU11 and PQMU12 can indicate that the voltage sag source locates between PQMU11 and PQMU12.

Compared Fig. 10(e) and (f) with Fig. 3(e) and (f), it is shown that the fault impedance would significantly affect instantaneous power. The duration of the integral should be prolonged when the high-impedance grounding faults happen. The simulation results show that the presented method can also properly indicate the location of the voltage sag sources.

4.2.5. Algorithm evaluation

Combining the simulation results, the following algorithm evaluations can be summarized.

- (1) The proposed algorithm is based on variation components resulting from the fault component superposition source, so that it is sensitive to voltage sag caused by the faults.
- (2) The inception angle of the faults and the fault impedance can affect the values of instantaneous energy based on variation components. In these situations the duration of the integral should be prolonged.
- (3) Capacitor operation will produce the associated disturbance energy, which can affect the value and direction of instantaneous energy based on variation components. In order to deal with the drawbacks of the proposed algorithm the suitable pick-up settings are necessary. And the proposed algorithm can be combined with the power quality disturbance recognition method to improve the performance of the proposed method.

5. Conclusions

This paper presents the investigation of using instantaneous active- and reactive-energy based on variation components for voltage sag source detection. The simulation studies have been performed in a 25 kV distribution system with DGs. The results show that the proposed algorithm can accurately detect the voltage sag source. Main conclusions of the method are drawn as follows:

- (1) The principle of the presented method is based on the variation components caused by the system faults, which are not affected by the different system operation mode.
- (2) The presented method is also effective in the single-source radial topology grid.
- (3) The fault type, the DG type, the inception angle of the fault and the fault impedance would affect the value of instantaneous active- and reactive-energy. However, the directions of instantaneous active- and reactive-energy can still distinctly show the voltage sag source location.

References

- [1] S. Santoso, E.J. Powers, W.M. Grady, A.C. Parsons, Power quality disturbance waveform recognition using wavelet-based neural classifier. I. Theoretical foundation, *IEEE Trans. Power Deliv.* 15 (2000) 222–228.
- [2] S. Santoso, E.J. Powers, W.M. Grady, A.C. Parsons, Power quality disturbance waveform recognition using wavelet-based neural classifier. II. Application, *IEEE Trans. Power Deliv.* 15 (2000) 229–235.
- [3] F. Zhao, R. Yang, Power quality disturbance recognition using S-transform, *IEEE Trans. Power Deliv.* 22 (2007) 944–950.
- [4] A.C. Parsons, W.M. Grady, E.J. Powers, J.C. Soward, A direction finder for power quality disturbances based upon disturbance power and energy, *IEEE Trans. Power Deliv.* 15 (2000) 1081–1086.
- [5] C. Li, T. Tayjasanant, W. Xu, X. Liu, Method for voltage-sag-source detection by investigating slope of the system trajectory, *IEE Proc. Gen. Transm. Distrib.* 150 (2003) 367–372.
- [6] N. Hamzah, A. Mohamed, A. Hussain, A new approach to locate the voltage sag source using real current component, *Electr. Power Syst. Res.* 72 (2004) 113–123.
- [7] A.K. Pradhan, A. Routray, Applying distance relay for voltage sag source detection, *IEEE Trans. Power Deliv.* 20 (2005) 529–531.
- [8] T. Tayjasanant, C. Li, W. Xu, A resistance sign-based method for voltage sag source detection, *IEEE Trans. Power Deliv.* 20 (2005) 2544–2551.
- [9] X. Dai, G. Liu, R. Gretsche, Generalized theory of instantaneous reactive quantity for multiphase power system, *IEEE Trans. Power Deliv.* 19 (2004) 965–972.
- [10] M. Saitou, T. Shimizu, Generalized theory of instantaneous active and reactive powers in single-phase circuits based on Hilbert transform, in: *Proceedings of the Power Electronics Specialists Conference, IEEE 33rd Annual, June 2002*, pp. 23–27.
- [11] Y. Xue, B. Xu, Z. Feng, Average and instantaneous reactive power of non-sinusoidal circuit based on Hilbert transformation, *Automat. Electr. Power Syst.* 28 (2004) 35–39 (in Chinese).
- [12] C.-C. Tseng, S.-C. Pei, Design and application discrete time fractional Hilbert transformer, *IEEE Trans. Circuits Syst. II: Analog Digital Signal Process.* 47 (2000) 1529–1533.
- [13] J. Hakan, W. Lars, Digital Hilbert transformers composed of identical all-pass sub-filters, in: *Proceedings of the 1998 IEEE International Symposium on Circuits and Systems, June 1998*, pp. 437–440.